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Chapter 20: The Ideology of Financial Market Pricing—Profit and Value in the Hidden Subspace

Abstract

The pricing system of financial markets is not merely a technical tool, but an institutionalized ideological apparatus. Using the Fiber Framework and the Macro-Cognitive Accounting Framework (MCAF) as diagnostic language, this chapter argues that shareholder value maximization (SVO), the Efficient Market Hypothesis (EMH), and the performativity of financial models together constitute the triple filter of the mapping function. Shareholder value maximization (SVO) determines the direction of the filter: only dimensions contributing to shareholder value enter the visible subspace. The ideological use of the Efficient Market Hypothesis (EMH) **suppresses** the legitimacy of questioning the filter itself: "the market is always right" removes system audit from the cognitive map. Performativity **narrows** the language for replacing the filter: model presuppositions are naturalized into common sense. The combined effect of the three is this: the mapping function of financial markets systematically encodes "dimensions that can be quarterly quantified" into the visible subspace, while pushing "dimensions that cannot be quarterly quantified" into the hidden subspace, thereby generating continuously accumulating cognitive debt, D_c. The empirical evidence comes from multiple levels: Lazonick's S&P 500 buyback data, 91% of profits flowing back to shareholders, and the latest trend of buybacks reaching approximately $1.02 trillion on a trailing-twelve-month basis in 2025; Asness's "less-efficient market hypothesis" raised from within the academic lineage of the EMH itself; MacKenzie's classic case-tracing of the performativ-

ity of Black-Scholes; and China's active dismantling of ideological apparatuses through the campaign to "break the doctrine of financial elitism." Within the transmission chain of the finance section, this chapter occupies the third link, moving from distorted incentives (Chapter 18) and the spatialization of the mapping chain (Chapter 19) toward institutional entropy increase and ideological masking, thereby connecting to the diagnosis of micro consequences in the next chapter.

Keywords

shareholder value maximization (SVO), Efficient Market Hypothesis (EMH), performativity, hidden subspace, pseudo-calibration, second derivative of narrative load (ϕ''), cognitive debt (D_c), buybacks, fiber inflation, ideological audit

§1 Introduction: Pricing Models Are Not Telescopes

Chapter 19 pushed the fiber diagnosis from the time window of a single market to the network structure of global continuous trading: night sessions, cross-border proxy pricing, 24-hour trading, and market fragmentation turned the mapping chain from linear to reticular, systematically diluting independent information components and generating proxy-pricing pseudo-small fibers. Those were chained problems across the L0-L3 layers of the mapping chain: the chain has become too reticular (L0: cross-market recursive references dilute independent components), too indirect (proxy pricing during closed-market hours contains no independent information), too dispersed (integrating scattered readings is itself a lossy mapping), and too self-referential (L3: the global financial cycle makes readings mutually determinative), making it difficult for the auditor to trace back. This chapter reaches a deeper layer: **the mapping chain itself is not ideologically neutral.**

The term "ideology" in this chapter refers to a conceptual framework that has been institutionalized and naturalized, and thereby exempted from systematic questioning. Its function is not to argue for a particular conclusion, but to define what counts as a "reasonable argument." In the case of the Efficient Market Hypothesis (EMH), it establishes "market prices are usually correct" as the default coordinate system, so that every deviation needs to be "explained," whereas the coordinate system itself need not be argued for. This is functionally equivalent to the second derivative of narrative load, ϕ'', in the Macro-Cognitive Accounting Framework (MCAF), with ϕ and its first-/second-order distinction defined in Chapter 3, §2.3, and ϕ'' positioned layer by layer within the four-layer progressive structure in Chapter 14, §2.

This chapter distinguishes shareholder financial value, the firm's long-term operating capability, and social value; the three do not always coincide. Omissions in pricing also have two distinct sources: in one, the relevant information has not been adequately identified; in the other, the relevant consequences, though already identified, are excluded from evaluation because they do not count toward shareholder returns. The former concerns information processing; the latter concerns evaluation goals and the attribution of costs.

The pricing instruments of financial markets—discounted cash flow (DCF) valuation, relative-valuation indicators exemplified by the price-earnings ratio (PE), the capital asset pricing model (CAPM), and Black-Scholes option pricing—are not merely tools that "describe how prices are formed," but also devices that "prescribe how prices ought to be computed." Each of them depends on specific choices of variables and conditions of applicability; when these instruments enter investment, assessment, and regulatory procedures, they help determine which information counts as valid evidence. In the language of the Fiber Framework, these instruments consti-

tute the **filter** of the mapping function, π: through variable selection and parameter setting, models foreground certain dimensions of the true state, T, and make other dimensions difficult to bring into evaluation, pushing them into the hidden subspace.

Evidence from within the financial industry itself—Asness's (2024) "Less-Efficient Market Hypothesis" and MacKenzie's "engine" thesis—supports this diagnosis, and is unfolded in §3 and §4 respectively.

This chapter argues the following proposition: **the pricing system of financial markets contains an institutionalized ideological apparatus, shareholder value maximization (SVO), the Efficient Market Hypothesis (EMH), and the performativity of financial models, which together constitute the filter of the mapping function and systematically encode "dimensions that can be quarterly quantified" into the visible subspace while pushing "dimensions that cannot be quarterly quantified" into the hidden subspace. This cannot be attributed to technical malfunction alone; it must also be traced to the institutional arrangements of evaluation goals, assessment rules, and the ways models are used.**

Relation to the surrounding chapters: Chapter 19 diagnosed the audit dilemmas brought about by the spatialization of the mapping chain—night-session access, proxy pricing, market fragmentation, and the recursive calibration of the global financial cycle, across the L0-L3 layers. This chapter shifts the diagnosis from market structure to the epistemic level, the institutional-lock aspect of pseudo-calibration at the L2 layer, asking how the mapping function itself is pre-filtered by ideological apparatuses. Within the transmission chain of the finance section, this chapter forms the third link: incentive distortion (Chapter 18) -> spatialization of the mapping chain (Chapter 19) -> institutional entropy increase and ideological masking (this chapter) -> micro consequences (Chapter 21) -> MCAF calibration (Chapter 22). The next chapter, Chapter 21, turns to concrete financial products and intermediary conduct: how three concrete financial products display different de-anchoring paths within the Fiber Framework, and how the key intermediary nodes along the investment/mapping chain shift, under the two pathways of structural compulsion and incentive induction, from auditors, exercisers of A, into amplifiers of narrative, producers of ϕ.

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§2 Shareholder Value Maximization: An Institutionally Reinforced Pricing Preference

§2.1 From Theoretical Claim to Governance Norm

The intellectual source of shareholder value maximization (SVO) can be traced back to Milton Friedman's 1970 article in *The New York Times Magazine*: "The Social Responsibility of Business Is to Increase Its Profits" (Friedman, 1970). In the 1970s this was still only a controversial theoretical claim. In 1976, Michael Jensen and William Meckling's agency-cost analysis provided the theoretical basis for designing managerial incentives around shareholder interests; in the 1980s the shareholder-value movement adopted it as its academic banner: managers are the "agents" of shareholders, and their goals should be subordinated to shareholder value maximization (Jensen & Meckling, 1976).

The key turning point occurred at the institutional level. Stiglitz summarizes the dominant orientation of U.S. corporate governance as shareholder value primacy (Stiglitz, 2019); this orientation has been continuously reinforced in legal precedent, capital-market discipline, and governance consensus, but whether it constitutes

a universal, singular, directly enforceable legal duty remains an important controversy in legal scholarship[1]. Whatever its legal status, this orientation is in practice no longer merely a "market preference" or "management philosophy": when a chief executive officer (CEO) invests resources in employee training, community development, or environmental protection rather than shareholder returns, he faces not only the market punishment of falling stock prices, but also potentially the pressure of shareholder litigation.

Within the Fiber Framework, the institutionalization of shareholder value maximization (SVO) in the United States means that the filter of the mapping function, π, is not spontaneously generated by the market, but **jointly reinforced and continuously consolidated by legal precedent, regulatory governance practice, and capital markets**. This institutional goal—maximize shareholder value—determines the orientation of the mapping chain. When employee training, users' long-term interests, and community relations cannot be converted, within the established assessment period, into confirmable shareholder returns, they tend to be undervalued in pricing evaluation—what is at work here includes not only the shareholder-value goal but also assessment horizons and evidentiary standards.

How this governance orientation affects the filtering of the mapping function must be examined through the concrete mechanisms of capital allocation and performance evaluation—stock buybacks provide one point of entry.

§2.2 Buybacks: Fabricating Calibration in the Visible Subspace

The institutionalization of shareholder value maximization (SVO) generates a concrete amplification mechanism: stock buybacks.

Lazonick's (2014) "Profits Without Prosperity" in *Harvard Business Review* provides the historical baseline: from 2003 to 2012, among 449 companies in the S&P 500 that remained continuously listed, 54% of net income was spent on stock buybacks and another 37% on dividend distribution. In total, an amount equivalent to 91% of net income flowed back to shareholders. Lazonick himself concluded that this left "very little for investments in productive capabilities or higher incomes for employees" (Lazonick, 2014). The ratio indicates the scale of shareholder distributions relative to net income; but net income is not a firm's entire source of investable funds—much R&D and capital expenditure is expensed before net income is computed, or is funded from operating cash flow—so one cannot infer from it that "only 9% remained for reinvestment."

Ten years later, the absolute scale of buybacks remains at a high level—but absolute amounts and distribution ratios are different statistical scopes, and any judgment of "acceleration" must be checked against each separately. Over the twelve months ending September 2025, S&P 500 buybacks totaled $1.02 trillion, up 11.1% year over year, a record for that rolling measure; this was the second time the measure surpassed $1 trillion (the first was the twelve months ending June 2022, at $1.005 trillion). S&P Dow Jones Indices at the same time projected that full-year 2025 buybacks would surpass $1 trillion for the first time on a calendar-year basis

[1]The status of shareholder value maximization (SVO) as "legal coercion" remains importantly contested in legal scholarship. Lynn Stout (2012), in *The Shareholder Value Myth*, argues that U.S. corporate law does not legally compel directors to maximize shareholder value. In *Dodge v. Ford* (1919), the core factual issue was that Ford, as controlling shareholder, refused to distribute dividends to minority shareholders; the essential meaning of the court's judgment was to protect minority shareholders from oppression by the controller, not to establish SVO as a universal legal duty. Stout's central claim is that shareholder value maximization is a widely believed "myth." That belief itself can produce real consequences by influencing behavior, but it is not a legal compulsion. Stout's analysis actually strengthens the diagnosis of the framework: if SVO is a widely believed "myth" rather than a legal obligation, then its self-fulfilling force, producing real consequences through shaping behavior, is precisely the core mechanism of an ideological apparatus. This chapter's diagnosis of the "institutionalization" of SVO therefore does not depend on the strong claim of "legal coercion."

(S&P Dow Jones Indices, 2025). A March 2026 MarketMinute market commentary, titled "The $1.2 Trillion Wall," points out that buyback amounts authorized by S&P 500 companies within 2026 will reach a record $1.2 trillion (MarketMinute, 2026b; this differs from the official S&P trailing-twelve-month figure of $1.02 trillion through Q3 2025 in statistical scope and release timing—the former is a forward-looking estimate of amounts authorized within 2026, the latter a rolling count of executed amounts). According to that commentary, since 2000 buybacks have accounted for nearly 100% of all net stock purchases in the U.S. equity market (MarketMinute, 2026b). The Magnificent 7 (the seven technology giants) account for roughly 30% of S&P 500 gross buyback spending (Goldman Sachs research, as relayed by Fortune, 2025). MarketMinute's commentary calls this phenomenon the "buyback engine"—"by aggressively reducing share counts, corporations are effectively 'manufacturing' value for remaining shareholders" (MarketMinute, 2026a). What is cited here is the narrative of a market commentary, not a validated research conclusion.

Within the Fiber Framework, the mechanism of buybacks can be diagnosed precisely. Earnings per share (EPS) is defined as:

$$\text{Basic EPS} = \frac{\text{Net income attributable to ordinary shareholders}}{\text{Weighted average of ordinary shares outstanding}}$$

Buybacks reduce the weighted average of shares outstanding used in the EPS calculation; when net income is positive and the buyback's financing cost and forgone income do not offset the denominator effect, EPS rises accordingly—an increase that does not equal a commensurate rise in operating profit. EPS is one of the most central readings in financial markets: valuation models (PE multiples), compensation incentives (stock-option exercise conditions), and analyst ratings (EPS beat or miss) are all directly tied to EPS. MarketMinute's commentary summarizes this mechanism as follows: by systematically retiring shares, nominal EPS growth can be raised even if net income is flat (MarketMinute, 2026a; a paraphrase based on that article, not a verbatim citation). The March 2026 commentary further points out that reliance on buybacks is "manufacturing" EPS growth—allowing mature large-cap companies to keep meeting analyst expectations even as organic revenue growth cools—and concealing the potential cooling of the global economy (MarketMinute, 2026b). In fiber language, "manufacturing EPS" is the precise description of pseudo-calibration at the institutional level: using a seemingly precise number, EPS growth, to conceal the true state, stagnating or declining organic revenue.

This is a specific form of pseudo-small fiber, δ_{pseudo}, for the concept see Chapter 4, §2.4: true fiber is large while perceived fiber is small. The precise number of EPS gives investors a sense of certainty, "this company earned $5.32 per share," but the uncertainty structure behind that number is concealed. What investors see is the upward trend of EPS; an investor who has not decomposed the increase into net-profit growth and share-count reduction may overlook how much of that increase comes from genuine profit growth and how much from a reduced share count. EPS can be computed accurately, yet by itself it cannot identify the sources of profit growth—pseudo-calibration consists in the interpretive misplacement that reads a per-share improvement produced by capital allocation as an improvement in operating capability, not in the division itself. The fiber δ_{EPS} is artificially compressed in the visible subspace, with the appearance of determinacy concealing the undecomposed state of the profit's sources, while it inflates in the hidden subspace—the firm's long-term reinvestment capacity, R&D spending, and human-capital accumulation are being systematically consumed.

But the 2025 data also reveal the structural tension in this model. Buybacks are highly concentrated among top firms: in Q2 2025 the top 20 companies accounted for 51.3% of total buyback volume, falling back to

49.5% in Q3, still above the historical average of 47.8% (S&P Dow Jones Indices, 2025). These data show that buyback spending is highly concentrated among a small number of large firms. The actual contribution of buyback concentration to index-level EPS growth must be computed separately from profit changes, share-count changes, and index weights, and cannot be inferred from concentration alone. In the same period, 17.1% of companies reduced the share counts used for EPS by at least 4% year over year (S&P Dow Jones Indices, 2025). When buyback concentration is superimposed upon the SVO filter, the distorting effect of a few firms' capital-allocation decisions upon the visible subspace at the index level is systematically magnified.

Lee, Shin, and Stulz (2016) found that growth opportunities proxied by industry Tobin's q (a market-valuation indicator) no longer persistently attracted more net funding after the mid-1990s—firms in high-q industries used excess cash to repurchase shares rather than raising more capital from the market. The study provides evidence on the relation between capital allocation and buybacks; but because growth opportunities are measured by a valuation indicator, it cannot by itself prove that real investment opportunities were systematically ignored—a measurement boundary that is itself a further instance of this chapter's §3 theme.

More direct micro-level evidence exists for repurchases conducted to meet analyst EPS forecasts: Almeida, Fos, and Kronlund (2016), using a regression discontinuity design, find that firms that would have just missed analyst forecasts without a repurchase are sharply more likely to conduct accretive buybacks, and that such EPS-motivated repurchases are followed over the next four quarters by reductions in employment, capital expenditure, and R&D—managers are willing to trade off investment and employment to meet analyst expectations. This evidence targets EPS-driven repurchases and cannot be extrapolated to all buybacks (Almeida, Fos, & Kronlund, 2016).

§2.3 Evidence from the Federal Reserve

Research by Gruber and Kamin (2017) provides empirical evidence from a cross-national perspective. In its analysis of multiple developed economies, the study finds that most countries have simultaneously experienced rising buybacks and dividends together with declining investment expenditure; the study also makes explicit that there is no statistically significant relationship between the two variables at the cross-national level—supporting neither the reading that rising buybacks cause reduced investment nor the reverse causation. The global coexistence of buyback expansion and investment slowdown is not a uniquely American phenomenon, but a phenomenal coexistence accompanying the global diffusion of the ideology of shareholder value maximization (SVO).

Within the Fiber Framework, this chapter treats this global coexistence as a phenomenon awaiting explanation: the diffusion of shareholder value maximization (SVO) as a governance norm alongside capital-market globalization is compatible with the coexistence of buyback expansion and investment slowdown—but this is an explanatory hypothesis, not a conclusion tested or demonstrated by the study above. The intuitive rebuttal to this hypothesis is: has the market itself not already developed environmental, social, and governance (ESG) evaluation frameworks to incorporate such dimensions? The analysis at the end of this section will show that the actual efficacy of the ESG attempt depends on whether these dimensions genuinely enter resource allocation and assessment—incorporation that remains at the level of discourse does not change the operation of the filter. If the hypothesis holds, the mapping functions of the whole world would prioritize dimensions that can be quarterly quantified, EPS, ROE, dividend yield, while systematically neglecting dimensions that cannot be quarterly quantified, R&D cycles, organizational capacity, environmental carrying capacity, and social safety

and fairness.

In 2025–2026, the global spread of shareholder value maximization (SVO) faces an important counter-narrative, stakeholder capitalism, but its subsequent trajectory revealed the gap between public commitments and governance constraints. The Business Roundtable's 2019 statement on "creating value for all stakeholders" was once regarded as marking the moment when stakeholder capitalism became the mainstream framework, but in retrospectives five years later, the chilling effect of political pressure and legal threats had already led many companies to abandon such commitments (Seegull, 2025). In 2025–2026, ESG in the United States encountered fierce political backlash, with Republican-led states passing a series of bills restricting ESG investment. The corporate response was not to defend stakeholder commitments, but to shift into "greenhushing," continuing the practice but no longer speaking openly about it (Bonime-Blanc, 2025). Seegull (2025) offers another explanation: managerial autonomy is expanding.

Within the Fiber Framework, stakeholder capitalism is not a replacement for SVO, but an adaptive deformation of SVO under new pressures of legitimacy. When ESG discourse remains at the level of the visible subspace, corporate annual reports and CEO letters, without changing the allocation of resources in the hidden subspace, it cannot serve as a substitute calibration. These materials present three distinct changes: some companies withdrew their commitments, some reduced public discourse while continuing the practices, and some managements used political pressure to expand their autonomy—evading both shareholder accountability and stakeholder accountability (Seegull, 2025). They cannot all be read as a one-way strengthening of SVO; the judgment they support is that the actual efficacy of stakeholder commitments depends on whether they enter resource allocation and governance procedures, not merely on public statements.

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§3 The Efficient Market Hypothesis: A Default Coordinate System That No Longer Needs Argument

§3.1 From Empirical Hypothesis to Ideological Apparatus

The Efficient Market Hypothesis (EMH) was proposed by Eugene Fama in the 1960s. Its core proposition is that asset prices "fully reflect" all available information at every moment (Fama, 1970). Originally, this was a testable empirical proposition: if markets are efficient, then trading strategies based on historical information should not systematically produce excess returns. What this chapter criticizes is not this academic proposition itself but its ideological use—the expansion of a conditional, jointly tested theoretical proposition into the institutional presumption that "market judgment needs no systematic audit"; "the market is always right" below refers to this expanded proposition.

But the fate of the EMH exceeded the domain of empirical science. It gradually deteriorated from a "testable hypothesis" into a "default coordinate system that no longer needs argument." Behavioral finance accumulated large quantities of anomalous evidence, momentum effects, value premia, overreaction, herd behavior. Testing market efficiency does face a joint-hypothesis problem (Fama, 1991): anomalous returns may indicate market inefficiency, but they may equally indicate an inappropriate risk benchmark. The problem lies not in researchers' revising risk models as such, but in whether these revisions are always used only to absorb counterexamples, never again exposing the efficiency proposition itself to the risk of revision or abandonment. The EMH became the benchmark model: every deviation had to be "explained," while the EMH itself no longer

needed explanation. The party that sets the benchmark also determines who must bear the burden of argument.

The completed form of this degeneration was publicly displayed at the Jackson Hole symposium on the eve of the crisis, in 2005. That symposium took as its theme a tribute to the soon-to-depart Greenspan era, while Rajan, then chief economist of the International Monetary Fund (IMF), submitted a paper, "Has Financial Development Made the World Riskier?," warning that the growing complexity of the financial system was accumulating tail risk. The author who raised a warning amid the tribute atmosphere was sharply criticized on the spot by several leading participants (Rajan, 2010); among them, Lawrence Summers judged the "basic, slightly Luddite premise" of the paper to be "largely misguided"[2], his argument being a transportation analogy—two centuries of transport evolution have made people increasingly dependent on intermediated arrangements; although the worst accidents are far larger in scale than ever before, transport fatalities have fallen dramatically, and overall welfare is overwhelmingly positive—financial innovation was classified as a continuation of the same process (Federal Reserve Bank of Kansas City, 2005).

This analogy displays an asymmetry in the burden of argument: the overall gains of innovation are placed in the default position, while risk warnings must first shake off the accusation of "resisting progress." The episode indicates a tendency of argument, but does not by itself confirm that its theoretical source is the Efficient Market Hypothesis (EMH). Three years later, the tail risks Rajan warned of materialized in concentrated form as a financial crisis—the specific prediction of an interbank-market freeze was vindicated within two years (Rajan, 2010).[3]

Mirowski (2013), in *Never Let a Serious Crisis Go to Waste*, points out directly that the Chicago School's Efficient Market Hypothesis is not only an academic theory, but a core component of neoliberal ideology. It provides the theoretical legitimacy for financial deregulation. If the market is always right, then regulation is an unnecessary intervention. After the 2008 financial crisis, the EMH was not abandoned; Mirowski (2013) documented and criticized an influential post-crisis attribution strategy—the crisis was attributed to government intervention, housing policy through the GSEs, rather than to systemic failure of the market itself—and this transfer of attribution strengthened the EMH's position at the ideological level. Akerlof and Shiller (2009), in *Animal Spirits*, provide a complementary diagnosis of this transfer of attribution: economic processes are jointly driven by five psychological elements, confidence, fairness, corruption and bad faith, money illusion, and stories (which they call "animal spirits"), and standard economics, by systematically excluding these elements from its baseline model, furnishes technical cover for the ideological presupposition that "the market is always right." This is structurally homologous with the second-derivative function of the EMH diagnosed in this chapter, the deletion of system audit from the cognitive map. It does not argue that the market is always right; it presupposes that "psychological elements do not belong to the class of legitimate analytic objects."

[2]Note on "Luddite." The Luddites were British handicraft workers who, between 1811 and 1816, smashed textile machinery in the name of the legendary "King Ludd," protesting against machines replacing human labor; the term was later generalized to refer to those who resist technological progress. Summers's phrase "slightly Luddite," in the original, modifies the premise of the paper rather than Rajan the person; its rhetorical effect is to pre-classify warnings about the risks of financial innovation under the stance of "resisting progress," rather than as an assessment of the substance of the risk itself.

[3]The methodological meaning of "degenerated into." The phrase "degenerated into" borrows from Imre Lakatos's methodology of scientific research programmes. A progressive research programme expands through predicting new facts; a degenerating programme absorbs anomalies by endlessly adjusting its protective belt. Behavioral-finance evidence has continued to accumulate, momentum, value premia, overreaction, herd behavior, while the hard core of the EMH, prices fully reflect information, has not been abandoned. Instead, anomalies have been explained away through the continual expansion of risk-factor lists, size, value, momentum, quality, profitability, and so on. Adding new risk factors does not by itself constitute degeneration; the judgment must further examine whether these revisions generate new testable content, rather than merely explaining existing results after the fact. This is not a challenge to the scholarly integrity of EMH supporters, but a methodological diagnosis of the structure of the programme itself.

In 2024, criticism emerging from within the academic lineage of the EMH itself provided the strongest evidence for this diagnosis. Cliff Asness, cofounder of the quantitative-investment giant AQR Capital Management and a doctoral student of Eugene Fama, published "The Less-Efficient Market Hypothesis." Asness holds that social media and a prolonged low-interest-rate environment distorted market information and detached stock prices from true value. His core judgment is that the informational efficiency of the market in the relative pricing of common stocks has become lower over the past thirty years, especially at medium-term timescales: "stock prices should accurately reflect reality," but in reality prices deviate from value more and more.

Within the Fiber Framework, Asness's diagnosis of "lower efficiency" can be read at three levels. **Level one (L1 optimization dynamics)**: viral dissemination on social media and the low-interest-rate environment reduce the cost of obtaining information, but also reduce the motivation to verify it. **Level two (L2 pseudo-calibration)**: the rise of index investing shifts the price-discovery function from active investors to passive capital, but passive capital does not perform fundamental valuation of individual stocks. **Level three (L3 recursive closure)**: when enough capital flows into indexes in a market-cap-weighted manner, rising prices of large firms automatically magnify their weight in the index, creating a self-reinforcing feedback loop. This is the positive feedback of passive investment already diagnosed in Chapter 17, §1.1.

Asness's conclusion is highly consistent with the ideological diagnosis of this chapter, but the explanation he offers remains more technical, social media, low interest rates, index investment. This chapter asks a deeper epistemic question: why can these technical factors continue to exist without being corrected by market forces? The answer is that the EMH itself functions as the ideology of a "default coordinate system." It classifies "market efficiency is declining" as an "anomaly in need of explanation," rather than as "systemic risk in need of audit." Asness concedes that corrective action remains possible—he explicitly invokes the "price is right" versus "no free lunch" dichotomy, noting that even where large deviations exist, exploiting them is harder and sticking with it is harder (Asness, 2024). What this chapter asks on top of that is why the supply of risk-bearing corrective capital is insufficient—the incentive structure of arbitrageurs, short-term ranking, benchmarking, and convex compensation, has been constrained by the institutional arrangements of shareholder value maximization (SVO) (Chapter 18). Shleifer and Vishny (1997), in "The Limits of Arbitrage," provide the theoretical basis of this mechanism: real-world professional arbitrage requires capital, bears risk, and uses other people's money. When prices deviate most severely from fundamentals, arbitrageurs also face the greatest pressure of capital withdrawal, and arbitrage is most likely to fail.

§3.2 In the Fiber Framework: The Operation of the Second Derivative of Narrative Load (ϕ)

Within the Fiber Framework, the ideological function of the Efficient Market Hypothesis (EMH) is a typical case of the second derivative of narrative load, ϕ, with ϕ defined in Chapter 3 as the narrative component attached to the mapping chain, affecting the acceptance and transmission of readings. The distinction between ϕ' and ϕ'' was given in Chapter 3, §2.3, applied to the AI domain in Chapter 9, and first positioned layer by layer within the four-layer progressive structure in Chapter 14, §2—this chapter applies that distinction to finance. From AI's "statistical skew in corpora" to finance's "presuppositions in pricing models," both share the same structure: the second-order narrative load suppresses the contrast condition required for audit. Shiller (2019), in *Narrative Economics*, supplies empirical calibration from the outside to this mechanism: economic events are not driven only by "fundamentals," but jointly driven by narratives that are contagious, mutable, and capable of remaining latent for long periods. Narrative itself is a constituent element of the economic causal chain, not merely rhetoric

added after the fact. This observation and this chapter's second-derivative diagnosis of ϕ divide the labor: the former discusses how narratives spread, the latter discusses which narratives gain institutional recognition. The ideological function of the EMH is precisely to lock critique at the ϕ' level inside "the anomaly list of behavioral finance," without allowing it to ascend to the ϕ'' level.

First-order narrative load, the first derivative of ϕ: specific market narratives affect investor behavior, "AI will change everything" drives up technology stocks, "inflation is about to spiral out of control" drives down bonds. These narratives can be identified and audited.

Second-order narrative load, the second derivative of ϕ: the EMH is not a specific market narrative, but **a meta-rule defining what counts as a 'reasonable narrative.'** Within the coordinate system of the EMH, "the market may systematically misprice" is not a legitimate narrative. It is classified as an "anomaly in need of explanation," not as "systemic risk in need of audit." What the EMH suppresses is not merely one particular contrast condition for audit, but **the legitimacy of the very concept of 'auditing the market itself.'**

If the market is always right, then "auditing whether market pricing itself is correct" becomes a meaningless question. Audit can only check whether "participants complied with the rules," compliance audit, but cannot ask whether "the rules themselves generate systemic error," system audit. The second-derivative function of the EMH is exactly this: **it deletes system audit from the cognitive map.**[4]

§3.3 Hayek's Price Mechanism and the Usurpation of the Efficient Market Hypothesis

Here a precise distinction must be made. Chapter 17, §2 cites Hayek's (1945) theory of the price mechanism: price is an aggregator of dispersed information, allowing economic activity to be coordinated without a central planner. This is a profound insight about **the efficiency of information transmission**.

The EMH commits a crucial usurpation in its inheritance of Hayek: Hayek says that "prices transmit information"; the EMH says that prices **always transmit information correctly**. The jump from "prices are useful signals" to "prices are always correct" is not a logical inference, but an ideological presupposition. Hayek himself was skeptical of the assumption of "complete rationality"; the EMH upgrades "the collective wisdom of market participants" into "the market cannot systematically make mistakes."

Within the Fiber Framework, Hayek's theory of the price mechanism says that "the mapping chain exists": the true state T can be partially transmitted through the price r. The ideological use of the EMH says that "the fiber of the mapping chain is a singleton"—price corresponds precisely to the true state, with no uncertainty. The Fiber Framework of Chapter 4 shows that fiber size must be interpreted under a specified state space and measure, and that a singleton fiber is a definitional limiting case (when the mapping function is fully known

[4]The place of the second derivative of ϕ within the formal system. In the meta-formula of the MCAF, ϕ is a scalar. The distinction between first and second derivatives is an empirical unfolding of ϕ in the financial domain. It reveals the internal structure of narrative load: ϕ', first-order narrative load, consists of concrete and auditable narratives, "housing prices always rise"; ϕ'', second-order narrative load, consists of meta-rules defining "what counts as a reasonable narrative," "market prices are always correct." The current version of the meta-formula does not yet unfold ϕ into a structured vector, (ϕ', ϕ''). This is first of all a strategic choice about framework maturity: introducing formal complexity too early, before sufficient empirical testing, may weaken the usability of the framework. But this does not mean that the internal structure of ϕ cannot be formalized in later versions. The current strategy is to make full diagnostic use of the distinction between ϕ' and ϕ'' (whose conceptual power has been demonstrated in Chapter 9's application to the AI domain and Chapter 14's layer-by-layer positioning). The prime notation in this chapter follows the hierarchical usage established since Chapter 9 and serves as a conceptual label; the original formulation of the first- and second-order distinction in Chapter 3, §2.3 is cast in the time dimension (the accelerating consolidation of $d^2\phi/dt^2 > 0$), and the hierarchical usage in this chapter, Chapter 9, and Chapter 14 is an extension of it—subsequent formalization will have to specify the variables and their relations separately before the two usages can be unified; at the formal level, the scalar form of ϕ in the meta-formula is for now left unchanged.

and the reading uniquely determines the truth, the fiber is a singleton—Chapter 4, §6); whereas in the multi-dimensional state-compression situations discussed in this book, a single price reading is far from sufficient to uniquely determine the complete true state. What the expanded proposition "prices are always right" evades is therefore this: prices cannot bear a complete state audit—this is precisely the Fiber Framework's rejection of it.

Asness's argument offers an even deeper contrast. Black's understanding of market efficiency was completely different from Fama's. In his presidential address to the American Finance Association, Black (1986) already systematically set out the central proposition of noise trading: noise makes markets "somewhat inefficient, but often prevents us from taking advantage of inefficiencies." Asness opens his paper with "stock prices should accurately reflect reality," but Black believed that price being exactly equal to value is impossible. Black's criterion is this: because all estimates of value are themselves noisy, "we can never know how far away price is from value"; he therefore redefined an efficient market as one in which "price is within a factor of 2 of value" (i.e., price is more than half of value and less than twice value), and on that basis held that "almost all markets are efficient almost all of the time"—where "almost all" means "at least 90%." This "factor of 2" criterion replaces the precise presupposition of "price equals value" with an interval several times wide.

Within the Fiber Framework, Black's position is equivalent to this: acknowledging, within a wide interval, the non-uniqueness of readings with respect to the true state—price can never correspond exactly to the true state, but can only fall within a wide interval. The EMH's upgrade of Hayek's "prices transmit information" into "prices are always correct" is precisely the expanded proposition that the Fiber Framework rejects. Asness's diagnosis of "less efficiency" is operationally valid, but the benchmark to which he refers, "the market used to be more efficient," may itself be an idealized retrospective construction. Black's position reminds us that at every historical moment market pricing is subject to the basic constraint that a single price reading is insufficient to uniquely determine the complete true state.

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§4 Performativity: How Financial Models Change from Description into Norm

§4.1 MacKenzie's Core Proposition

In *An Engine, Not a Camera* (2006), Donald MacKenzie puts forward a proposition that changed economic sociology: financial models do not passively describe markets, the "camera," but actively shape them, the "engine." Chapter 6, §5.2 has already cited MacKenzie's "engine, not a camera" proposition to analyze the monopoly of the mapping function, a question of power. This chapter turns the analysis to the presuppositions of the mapping function itself: from "who controls the mapping function" to "what the mapping function itself presupposes," an epistemic question. The edited volume *Do Economists Make Markets?* (2007), compiled by MacKenzie, Muniesa, and Siu, extends the "engine, not a camera" proposition from the single Black-Scholes case into a cross-domain collection. Its core proposition—that economics does not passively observe how the economy works, but performs, shapes, and formats it—was first systematically advanced by Callon (1998) in the introduction to *The Laws of the Markets*: "economics, in the broad sense of the term, performs, shapes and formats the economy, rather than observing how it functions"; MacKenzie (2006) systematically unfolded it in financial markets through the Black-Scholes case, and the 2007 volume gathered the discussion into further domains beyond finance; the EMH and index-tracking funds are listed as important practical carriers of this performative mechanism. These cases from different domains show that the phenomenon of models shaping

markets is not limited to Black-Scholes; whether a model produces such an effect, and in what direction, still depends on how it is used and on the institutional and technological conditions.

MacKenzie's classic case is the Black-Scholes option-pricing formula. When Fischer Black and Myron Scholes published the formula in 1973, the Chicago Board Options Exchange had just opened, and option traders still priced mainly through intuition and experience. At first, market prices did not perfectly coincide with the Black-Scholes model: market prices deviated from the model's predictions. But as more and more traders used Black-Scholes to calculate the "theoretical price," market prices began to converge toward the model's predictions. The model "created" the reality it described.

MacKenzie calls this phenomenon "Barnesian performativity," named after sociologist of science Barry Barnes and his emphasis on the central role of self-validating feedback loops in social life. The performativity of financial models is not an accidental side effect, but the systematic consequence of widespread adoption: when enough participants use the same model for pricing, the model becomes the participants' common "anchor," and market prices converge toward the model's forecast. MacKenzie deliberately does not equate the phenomenon with the "self-fulfilling prophecy" in the sense of Robert K. Merton—the latter refers to false or arbitrary beliefs that come true because they are widely believed, whereas an arbitrary mispricing formula, even if widely adopted, would be abandoned because it generates systematic losses and arbitrage opportunities; what can become "truer" through use is precisely a model with genuine methodological content.

MacKenzie's study of high-frequency trading in *Trading at the Speed of Light* systematically unfolds this proposition: the trading behavior of algorithms shapes market microstructure—in "market-impact trading," algorithms profit from the anticipated impact of their own behavior on other algorithms; interactions among algorithms within the same market-making firm produce emergent consequences that no one designed—a self-trade prevention system unexpectedly raised the profitability of market-making algorithms; in the 2012 Knight Capital incident, a faulty algorithm accumulated a $6.65 billion position within 45 minutes, causing a $460 million loss—human employees took the same three-quarters of an hour to locate and shut down the faulty system (MacKenzie classifies the case as one of technical-system failure; this chapter takes its aspect of "algorithmic collective behavior altering the market environment") (MacKenzie, 2021). He reaffirmed the "engine" proposition in a 2025 podcast interview (MacKenzie, 2025). Within the Fiber Framework, this is the concrete unfolding of performativity in the dimension of speed: algorithms are not merely "fast"; their collective behavior creates a new market reality, while the process by which that reality is generated is difficult to track in real time.

§4.2 The Second-Order Effect of Performativity: The Naturalization of Presuppositions

MacKenzie's analysis also reveals a deeper effect of performativity: **the naturalization of model presuppositions**.

The Black-Scholes model depends on a strict set of assumptions: continuous trading, frictionless markets, constant volatility, and a stock price that follows geometric Brownian motion—under which future prices are log-normally distributed and log returns are normally distributed. When the model was first introduced, traders clearly understood that these were "simplifying assumptions"; they knew that volatility was not constant and that the tails of actual returns are far fatter than the model's normal assumption allows. But as the model became the industry standard, these assumptions gradually ceased to be questioned as "assumptions" and were instead accepted as "the nature of the market." Traders began to use "implied volatility" as basic language, a concept

that makes sense only within the Black-Scholes framework.

Before the 1987 crash, the implied-volatility surface of S&P 500 options was flat, consistent with Black-Scholes's assumption of constant volatility. After the crash, the surface took on a "skewed" shape—the implied volatility of deep out-of-the-money put options was significantly higher than that of at-the-money options (MacKenzie's 2006 original text records the pattern as "volatility skew or smile"; in market convention, the one-sided asymmetric form that index options took on after the crash is usually called a skew, to distinguish it from the symmetric smile with both ends elevated—this chapter follows the market-convention usage). Traders called this the "volatility skew," but the "volatility skew" is itself a concept internal to the Black-Scholes framework. Outside Black-Scholes, "implied volatility" and "volatility skew" have no meaning. The model failed, but the language of the model remained the only language available for discussion.

Within the Fiber Framework, this is the operation of the second derivative of ϕ at the level of pricing models: Black-Scholes not only changed prices, the first-order effect, but also changed the **conceptual framework through which traders understand prices**, the second-order effect. Once everyone thinks in terms of "implied volatility," alternative frameworks bear an additional expressive cost—once a model has become the universal tool, its concepts may outlast its original assumptions; traders can know that the constant-volatility assumption does not hold and still use Black-Scholes implied volatility to compare and express option prices. What forms here is a path dependence in quoting and communication, not a full commitment to the model's assumptions. What this chapter attends to is this asymmetry, not the claim that model users can no longer criticize the model.

Boundary marking: MacKenzie's theory of performativity contains a richer spectrum than what is presented in this chapter. **Barnesian performativity**, named after Barry Barnes, means that widespread use of a model makes the market move closer to the model's prediction, as in the early history of Black-Scholes. **Counterperformativity** means that the use of the model makes the market deviate further from the model's prediction—for example, if portfolio-insurance trading grounded in option theory amplified price declines, such a reverse feedback could arise. The emergence of the volatility skew is first of all evidence of model misfit, and counterperformativity cannot be confirmed merely by its existence. This chapter focuses on Barnesian performativity because within the triple apparatus of SVO-EMH-performativity it plays the role of **operational solidification**, naturalizing model presuppositions into common sense. But the existence of counterperformativity marks an important boundary: the control exercised by ideological apparatuses is not absolute, and the market can resist the model. This creates a productive tension with the EMH's degeneration into the default coordinate system. Why are some models domesticated by the market, option prices converging to Black-Scholes, while others encounter market resistance, as in the volatility skew? This tension itself is a phenomenon that the Fiber Framework will need to explain in the future.

It also needs to be marked that **within the quantitative-finance practitioner community there exists an independent critical tradition**, which observes the same class of phenomena as the sociological diagnosis of performativity. Around the problem of the "naturalization of presuppositions" diagnosed in §4.2, this tradition converges on three propositions: **models are metaphors, not theories** (Derman, 2011), financial models have no "truth" in the sense of physical law, but are approximate analogical tools, though the market often uses them as if they were theories; **backtest overfitting is arguably the most important open problem in mathematical finance—its "P versus NP"** (López de Prado, 2018), in a sufficiently large strategy space with sufficiently rich historical data one can always find strategies that appear statistically significant, while a single backtest cannot reliably distinguish real skill from overfitting ex ante; and **markets are shaped into the image described by**

the model (MacKenzie, 2006, the external diagnosis of economic sociology), performativity is not an extra state of the model but the structural consequence of widespread market adoption of the model. These three propositions converge at the epistemological level, the ontology of the model; the methodological level, the difficulty of distinguishing overfitting from true skill; and the sociological level, the feedback effect of the model upon the market, toward one common direction: the epistemic status of financial pricing models differs from that of theory in natural science.[5] The first two propositions are the internal testimony of practitioners about the limits of the tool, the third is the external diagnosis of the structural consequences of those limits; the three are mutually illuminating rather than parts of a single tradition.

§4.3 Echoes of Chapter 19: The Fiber of VIX and Implied Volatility

The previous chapter, Chapter 19, §5.2, analyzed the fiber properties of the volatility index (VIX) as a proxy variable for global risk appetite: VIX's methodology is built on a model-free continuum-of-strikes framework, and when markets fluctuate violently and the quotation continuity and liquidity of deep out-of-the-money options deteriorate simultaneously, the "continuum of strikes" assumption effectively fails and the set of eligible options drifts, so that the fiber of VIX inflates sharply (the diagnosis and its methodological citations are given in Chapter 19, §5.2). MacKenzie's theory of performativity explains a further layer: although the current VIX calculation no longer depends on the Black-Scholes constant-volatility assumption, the market to this day uses the vocabulary left behind by the option-pricing framework—"implied volatility," "volatility skew"—to quote, compare, and talk about options. As the institutionalized product of this conceptual language, VIX acquired the authority of "measuring risk"; yet its "precise reading" conceals two layers of dependence: it conceals the engineering dependence on quotation continuity, and it conceals the proxy extrapolation from S&P 500 option readings to "global risk appetite." This is the operation of the pseudo-small fiber at the level of the meta-reading.

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§5 The Institutionalized Production of the Hidden Subspace

§5.1 The Superposition of the Triple Apparatus

Shareholder value maximization (SVO), the Efficient Market Hypothesis (EMH), and performativity are not three ideological mechanisms of the same kind, but a **superposed institutional apparatus**: SVO is a governance goal; the EMH is a theoretical hypothesis about the relation between prices and information; performativity describes a feedback between knowledge tools and practice. The apparatus is formed by the specific conjunction of the three—the governance goal shapes the choice of metrics, the ideological use of the efficiency hypothesis raises the threshold for questioning the metrics, and the universal model and its operating procedures make the established metrics harder to replace. Such a conjunction can reinforce the prevailing evaluative order, but it cannot be derived from any single link alone.

SVO determines the **institutional goal** of the mapping chain: maximize shareholder value. This determines which dimensions are encoded, only dimensions contributing to shareholder value enter the visible subspace.

The EMH determines the **epistemic premise** of the mapping chain: market prices are always correct. This

[5] Related discussions can also be found in Wilmott and Orrell (2017), whose diagnosis is consonant with the three propositions above but more focused on risk models; Bookstaber (2007, *A Demon of Our Own Design*; 2017, *The End of Theory*) analyzes the limits of models and their consequences from the perspective of financial innovation and complex systems.

suppresses the legitimacy of "system audit." If the market is always right, then questioning the logic of pricing itself becomes unnecessary.

Performativity realizes the **operational solidification** of the mapping chain: pricing models change from "description" into "norm," and the model's presuppositions are naturalized into "common sense." Even if individual participants doubt the correctness of the model, they cannot step outside the model's language in order to articulate their doubt.

The combined effect of the three is this: SVO determines the **direction** of the filter, toward shareholder value; the EMH **suppresses** the **legitimacy** of questioning the filter, the market is always right; and performativity **narrows** the **language** of replacing the filter, the conceptual framework of the model becomes the only conceptual framework.

§5.2 The Concrete Contents of the Visible Subspace and the Hidden Subspace

Under this triple apparatus, the mapping function, π, of the financial market systematically allocates different dimensions of the true state, T, into the visible subspace and the hidden subspace:

Visible subspace (dimensions encoded by pricing models): - EPS and its growth rate - PE and PB - Dividend yield and buyback scale - Quarterly revenue and profit margin - Free cash flow and return on invested capital (ROIC) - Credit ratings and debt ratios - Implied volatility and option Greeks

These dimensions share one common feature: **they can be quarterly quantified, model-calculated, compared, and ranked**.

Hidden subspace (dimensions systematically ignored by pricing models): - employee skill accumulation and organizational learning curves - the long-term return on R&D spending, with realization horizons varying by industry, project, and technology route - supply-chain resilience, parts of whose vulnerability go insufficiently recognized in routine evaluation and are exposed in concentrated form when shocks arrive - environmental externalities and resource depletion - community relationships and social license to operate - management's cognitive quality and long-term strategic capacity - institutional trust and legitimacy base, $\|L\|$, the concept defined in Chapter 3 as the level of the legitimacy base supporting institutional operation, determined by its weakest dimension

These dimensions also share one common feature: **they cannot be quarterly quantified, cannot be calculated by standard models, and cannot be simply compared and ranked**. A note is needed: the two-column distinction above describes relative visibility within evaluation, not whether the dimensions can in principle be quantified—the same dimension can partly enter indicators while retaining content that is hard to verify or left unaccounted.[6]

§5.3 The Amplification Mechanism of the "Virtual" and the "Real"

The institutional production of the hidden subspace generates a particular amplification mechanism of the "virtual" and the "real." There are two paths for profit maximization:

[6]This chapter uses the "quarter" to mark an institutionalized evaluation cycle, not to treat it as a natural epistemic boundary. The specific establishment of the U.S. quarterly-report system (10-Q) is a detail of institutional history and is not unfolded here; the problem lies not in any fixed unit of time, but in the structural mismatch between the assessment window and the time required for capability formation and risk exposure; the disclosure cycle and managers' actual assessment horizons should also be distinguished.

The "real" path: increasing the firm's true earning capacity by creating real value, developing new products, improving productive efficiency, and expanding into new markets. On this path, the change in readings, rising EPS, corresponds to a change in the true state, T, and the fiber size remains within a normal range.

The "virtual" path: taking the improvement of evaluation metrics as its direct goal—arranging buybacks around EPS targets, improperly capitalizing expenditures, improperly recognizing revenue early, and deliberately lowering guidance in order to "beat expectations." This path does not leave the firm's true state completely unchanged—buybacks alter cash and equity structure—but it does not presuppose any commensurate improvement in operating capability; the change in readings, rising EPS, has no corresponding change in operating capability, and the fiber is artificially compressed in the visible subspace while silently inflating in the hidden subspace. Buybacks are not in themselves pseudo-calibration; they enter the mechanism discussed in this chapter only when the improvement in per-share metrics is taken as an improvement in operating capability, concealing their financing cost or opportunity cost.

The institutional reasons why SVO prefers the "virtual" path are clear:

First, asymmetry of time preference. The "virtual" path works within the current quarter, buybacks immediately raise EPS. The "real" path takes years to work, with the return period on R&D investment varying by industry and project. The quarterly-report system forces managers to display "progress" within a three-month window, and buybacks are among the few operations that can directly alter the EPS denominator within three months—their effect usually shows up faster than long-term investment, but not unconditionally: when net income is negative, a reduction in share count deepens the per-share loss instead.

Second, asymmetry of risk. The "virtual" path has low risk, because in the common case of positive net income the effect of buybacks on the denominator is nearly deterministic: reducing the weighted average share count raises EPS. The "real" path has high risk: R&D may fail, new markets may not exist, and organizational reform may be resisted. Under the SVO framework, management compensation is tied to share price, as analyzed in Chapter 18, §1, so choosing the path with the more certain result is rational, see especially Chapter 18, §1.5, where the compression of multidimensional states into a one-dimensional signal makes the hidden costs of the "virtual" path invisible to decision makers.

Third, asymmetry of observability. The "virtual" path immediately produces a signal in the visible subspace, EPS rises, analysts upgrade ratings, share price rises. The "real" path may produce a negative signal in the visible subspace in the short term, R&D spending rises and margins fall, analysts downgrade the stock.

§5.4 The Financial Correspondent of Cognitive Debt (D_c)

In the language of the Macro-Cognitive Accounting Framework (MCAF), the inflation of the hidden subspace corresponds to the accumulation of cognitive debt, D_c, the total accumulated gap between readings and the true state.

When firms choose the "virtual" path, the gap between the visible readings, EPS and share price, and the true state, T, widens. In the situations discussed in this chapter, this gap takes the form of a persistent misplacement of evaluation: financial readings are taken as sufficient evidence of operating capability. The Fiber Framework serves to reveal that readings are insufficient to identify the true state; the concept of cognitive debt describes the consequences of this insufficiency remaining uncorrected over time—the two are related, but one cannot infer a widening fiber or growing debt merely from rising readings. Because the misplacement occurs in the hidden

subspace, what participants observe is "everything is normal," or even "conditions are improving," because EPS is rising. This is the core mechanism of cognitive debt: **visible readings improve, while the invisible true state deteriorates, and the gap between the two is cognitive debt**.

Lazonick's (2014) distribution data provide one point of entry for observing the accumulation of cognitive debt: shareholder distributions occupied an important position in the capital allocation of the sample firms. Such aggregate ratios cannot by themselves prove a decline in long-term capability; what they prompt the auditor to check are two things—whether firms cut necessary investment in order to meet current-period targets, and whether such cuts were concealed by favorable financial readings. This chapter locates cognitive debt in that misplacement—the gap between improving visible readings and deteriorating true capability—rather than in any number that can be read off directly from a distribution ratio.

Cognitive debt is usually liquidated suddenly. Once deterioration in the hidden subspace accumulates to a certain degree, it breaks through into the visible subspace in the form of the "unexpected": the sudden rupture of supply chains (as in the 2020 pandemic); core technology suddenly surpassed by competitors; environmental disaster suddenly exposed. External shocks may bring to light vulnerabilities that have long gone unrecognized—here the trigger must be distinguished from the mechanism of loss formation: the shock itself is not produced by cognitive debt, but its consequences may be aggravated by long-standing omissions. And not all cognitive debt erupts in black-swan form—cognitive debt may also be realized gradually through slow declines in competitiveness, sustained losses, or the intergenerational transfer of costs, rather than through a single crisis. Fiber has long inflated in the hidden subspace, but it is only "discovered" once it crosses into the visible subspace. Kindleberger and Aliber (2015), in their four-hundred-year history of financial crises, provide cross-cycle historical evidence for this mechanism of "concentrated manifestation": the timing of crises is often unpredictable, but the credit expansion, asset-price deviation, and leverage accumulation preceding them are traceable. The "surprise" lies in the moment, not in the mechanism. The book continues Minsky's line of inquiry, as the continuation editor Aliber explicitly points out in the introduction, and provides a historical reference for this chapter's distinction.

§5.5 International Spillover: The Washington Consensus and the International Institutionalization of the Ideological Filter

§5.1–§5.4 dealt with the institutionalized production of the hidden subspace within a single country—the triple superposition of shareholder value maximization (SVO), the Efficient Market Hypothesis (EMH), and performativity makes certain policies routine options, while others must additionally justify their legitimacy. But the same filtering mechanism does not stop at national borders. This section discusses its extension at the level of international institutions: the Washington Consensus.

The Content and Institutional Carriers of the Washington Consensus. Williamson (1990), in *Latin American Adjustment: How Much Has Happened?*, was the first to give a set of policy prescriptions the systematic name "Washington Consensus," positioning it as ten policy instruments on which Washington in 1989—the international financial institutions (the IMF and the World Bank), the economic agencies of the U.S. government (with the Treasury foremost), the Federal Reserve Board, and the think tanks—had formed a considerable degree of consensus regarding their correct application, in response to Latin American reform problems. The core ten items include: fiscal discipline; redirecting public expenditure toward education, health (especially parts benefiting the disadvantaged), and infrastructure investment; broadening the tax base while maintaining moderate

marginal tax rates; market-determined interest rates with positive real rates; maintaining a competitive exchange rate; trade liberalization; opening to foreign direct investment; privatization of state-owned enterprises; reducing regulations that impede market access and competition; and securing property rights (Williamson, 1990).

A marking is required: the foreign-investment opening discussed in Williamson's original text was limited to foreign direct investment, and the original text explicitly states that the liberalization of foreign financial flows "is not regarded as a high priority." Attributing full capital-account liberalization to the "Washington Consensus" belongs to the broadened policy paradigm later attached to that name; the discussion of the Asian financial crisis below targets this broadened paradigm, not Williamson's original ten items.

The institutional carriers are IMF/World Bank conditionality lending, the sovereign-rating methodologies of credit-rating agencies, the U.S. Treasury's bilateral negotiating positions, and the academic training that mainstream economics doctoral programs provide to policy officials from developing countries.

The Operating Mechanism as an L2 Ideological Filter. The Washington Consensus is not a financial model—it has no pricing formula and does not directly generate price readings. But it performs a filtering function isomorphic with §3 (the Efficient Market Hypothesis) and §4 (performativity): it partitions the national policy option space into a visible subspace (openness, liberalization, discipline) and a hidden subspace (capital controls, industrial policy, directed credit, state-owned finance). This partition does not rely on legal coercion; it is accomplished through the multiple superposition of the policy conditionality attached to loans, the implicit weights of rating methodologies, and the terminology base of IMF Article IV consultation assessments; how academic training and journal-publication mechanisms participate in the reproduction of this normative pressure is a channel that still requires separate examination. The result: openness becomes the default baseline requiring no argument ("sound economic common sense"), while control becomes the anomalous state that must be argued for ("distortion," "politicization," "breeding ground for rent-seeking"). In the language of fiber, this is the international-policy-level concretization of the pseudo-small fiber described in §6.1—the assessment readings read by policymakers in developing countries (ratings, loan conditions, IMF report language) are all "precise," but the structure of uncertainty behind those readings—which policy dimensions have been excluded from assessment in advance—is invisible to the decision-makers themselves.

The 1997 Asian Financial Crisis as a Stress Test. Before the 1997 crisis, Thailand, Indonesia, and South Korea advanced financial and capital-flow liberalization to different degrees and in different sequences—Stiglitz notes that this liberalization was persistently pushed by the IMF and the U.S. Treasury in the late 1980s and early 1990s, and regards it as probably the single most important factor leading to the crisis (Stiglitz, 2002); after the crisis broke out, the standard prescription the IMF issued was still fiscal austerity, higher interest rates, and further opening. These prescriptions are logically consistent within the standard Washington Consensus framework, because the framework's visible subspace simply does not contain the option that "capital controls may be a legitimate emergency instrument." After Malaysia's Mahathir imposed capital controls, his policy was attacked from all sides—the semantic structure of this characterization is fully isomorphic with §3's Efficient Market Hypothesis marking "market inefficiency" as an anomalous state requiring argument: the verdict requires no argument, because the framework has already presupposed it.

Stiglitz (2002, *Globalization and Its Discontents*), as the World Bank's chief economist during the crisis, offered a systematic critique: the IMF's failure was not technical misjudgment but structural—its analytical framework was ideologically preloaded, systematically unable to see reality inconsistent with the preloaded framework.

In this chapter's diagnosis, Stiglitz's thesis corresponds to the international-institution-level form of "the auditor being itself embedded in the ideological apparatus" described in §6.3—in the standard competitive model in which IMF economists were trained, what Stiglitz calls "the model that underlies the IMF's market fundamentalism," demand always equals supply and involuntary unemployment does not exist; this toolkit has itself encoded the presupposition that "markets are spontaneously effective," making it difficult for them to detect with it the blind spots of the tool itself.

Soros's Dual Position. Soros (2002, *George Soros on Globalization*) criticized the Washington Consensus from a particular position—he is at once one of the greatest beneficiaries of international capital markets and one of the keenest observers of their structural consequences. His core proposition: "the visible hand is that of the United States, supported by the Washington consensus"—financial markets do not tend toward equilibrium; they need a visible hand to guide them and keep them from going off the rails; and today that visible hand is the United States, with the Washington consensus serving as its institutional support (Soros, 2002).

From this chapter's L2 perspective, this proposition admits a further layer: the Washington Consensus not only provides institutional support for this dominant position (as Soros emphasizes), it also naturalizes a set of financial-discipline ideology into "universal economic common sense"—so that policies at odds with it must first bear the burden of argument, while conforming ones need not. This is precisely the second-order operation of the ideological filter described in §6.3—it not only affects behavior, but also presets the conceptual framework of the auditors.

The Division of Labor with Chapter 17, §6.5. Chapter 17, §6.5 analyzed the institutional consequences of the Washington Consensus from the perspective of the structural stability of the master clock (the U.S. dollar) as a recursive anchor—how the master clock's influence squeezes the policy space of developing countries through conditionality lending and rating mechanisms (an L3-level recursive-closure diagnosis). This section's perspective is complementary but not duplicative: not the institutional consequences, but how the filter is installed into the conceptual framework of decision-makers (L2-level ideological filtering). The confluence of the two perspectives yields a more complete diagnosis: the master clock's international influence and the ideological filter support each other—the master clock's network effects make "accepting the Washington Consensus" a tacit condition for access to international liquidity, while the Washington Consensus's ideological function in turn naturalizes this acceptance into "economic common sense," leading decision-makers to take a contestable policy choice as settled common sense.

Boundary Marking and Open Questions. The domestic reverse calibration of the hidden subspace (§5.6) is backed by concrete political authority and institutional pathways. But reverse calibration at the international level has no equivalent executive subject—no single political authority can declare any principle of the Washington Consensus an "erroneous idea." Mirowski's (2013) diagnosis points to the survival path of neoliberalism after the 2008 crisis—it achieved self-perpetuation by attributing the crisis itself to "excessive government intervention rather than market failure"; this diagnosis applies equally to the Washington Consensus: the post-1997 discussion (that capital controls may be reasonable) was absorbed as "a second-best instrument to be considered in specific circumstances"; the international policy framework has since undergone adjustments that entered formal policy: the IMF adopted an institutional view on the liberalization and management of capital flows in 2012, recognizing that capital flow management measures can be useful in certain circumstances but should not substitute for warranted macroeconomic adjustment, and its 2022 review further allowed for the preemptive use of CFM/MPMs—measures that are both capital flow management and macroprudential—under certain vulner-

ability conditions (IMF, 2012; IMF, 2022). Whether these adjustments changed the basic presupposition of openness-first, or mainly modified its conditions of application, still requires concrete analysis and cannot be presumed to have left the framework unchanged. That the L2 ideological filter has deeper structural inertia at the international level than at the domestic level is an open problem for which this framework does not yet provide a clear dismantling path.

§5.6 Reverse Calibration: "Breaking the Doctrine of Financial Elitism" and the Active Dismantling of Ideological Apparatuses

The inflation of the hidden subspace eventually breaks into the visible subspace in the form of the "unexpected," but when it does, whether it is used as an opportunity to reinforce the existing ideology, as after the 2008 crisis when it was attributed to government intervention, or as an occasion to actively dismantle the ideological apparatus depends on the political conditions and institutional forces of the time. The triple apparatus is not something that can only be diagnosed; it can also be actively dismantled. In 2023, the website of the Central Commission for Discipline Inspection and the National Supervisory Commission published an article by its Thirteenth Review and Investigation Office explicitly calling for breaking such erroneous ideas as "the doctrine of financial elitism," "money-only doctrine," and "aligning with the West" (Central Commission for Discipline Inspection and the National Supervisory Commission, Thirteenth Review and Investigation Office, 2023); one month later, in summarizing the 2020–2022 three-year action on corporate governance, the China Banking and Insurance Regulatory Commission likewise called for breaking the "doctrine of financial elitism" and encouraged the exploration of non-material incentives (China Banking and Insurance Regulatory Commission, 2023)—these statements echo the compensation-restraint institutions already established in bulk; the performance-based pay clawback mechanism had been built out across the banking and insurance sectors since the 2021 guidance document, covering more than 95% of institutions by 2023 (Chapter 18; China Banking and Insurance Regulatory Commission, 2023).

Within the Fiber Framework, this constitutes an institutional counterattack against the fourth link of the "drifting from the real into the virtual" cross-layer transmission chain (ideological masking) (the "drifting from the real into the virtual" four-link chain and the finance-section five-link transmission chain set out in §1 of this chapter are two distinct structures; see Chapter 17, §0 and its footnote 2). The target is not one particular compensation number or regulatory rule, but the meta-rules defining "what counts as reasonable." Narratives that had previously been naturalized, such as "high compensation in finance is the natural outcome of the talent market" and "financial practitioners are elites," were directly classified by the official discourse as "erroneous ideas."

This ideological calibration further affects incentive arrangements, regulatory institutions, and product evaluation. At the incentive layer: once the "elitist narrative" is broken, the existing pay-clawback institutions gain stronger backing from political authority—the institutions themselves had already moved from "policy advocacy" to "industry standard" with the 2021 guidance document (Chapter 18), and the ideological demotion provides them with further normative support. At the institutional layer: the narrative that "financial innovation requires freedom" is weakened, and the writing of look-through supervision into law no longer faces the same ideological resistance as "excessive intervention" (Chapter 18). At the product layer: these official statements provide normative support for questioning the value of complex products—whether complexity no longer carries a legitimacy premium, and whether investor behavior has in fact changed, must still be judged from implementation materials. The common effect of the three statements is that ideological calibration and

the existing institutional arrangements echo one another; whether this constitutes actual calibration still depends on whether the neglected dimensions can enter evaluation and whether evaluation outcomes are open to review (see the boundary marking below).

Boundary marking: "breaking the doctrine of financial elitism" is a political signal, not an academic argument, and its effectiveness depends on the continuing input of official power. Minsky's (1986/2008) mechanism that "stability breeds instability" applies to ideological calibration as well. Once crisis memory fades, whether the "elitist doctrine" will, like financial regulation, face periodic pressure toward relaxation is a question that cannot be answered at present, but must be marked. Minsky originally diagnosed the endogenous instability of financial structures themselves, the cyclical evolution from hedge finance to speculative finance to Ponzi finance. This chapter applies that mechanism analogically at the ideological level: the steady state of ideological calibration may likewise breed the seeds of its own loosening within stability.

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§6 Fiber Diagnosis: Ideology as the L2 Layer of the Mapping Function

§6.1 Position within the Four-Layer Progressive Structure

Chapter 19 used the four-layer progressive structure (L0-L3, Chapter 4, §7) to complete its comprehensive diagnosis of the spatialization of the mapping chain: night-session access and 24-hour dispersed pricing fall at the L0 layer (the mapping chain turns from linear to reticular); the redistribution of access rights in price formation and high-frequency trading fall at the L1 layer (optimization dynamics); proxy-pricing pseudo-small fibers and NBBO integration fall at the L2 layer as market-structure forms; and the mutual calibration of the global financial cycle ascends to the L3 layer (recursive closure). But the L2 diagnosis there concerns pseudo-calibration of the market structure itself; the ideological apparatus diagnosed in this chapter corresponds to the other form of the L2 layer—the **institutional-lock aspect: pseudo-calibration** (the concept defined in Chapter 4, §2.4: the system appears to be performing calibration, but is actually manufacturing pseudo-small fibers).

The superposition of SVO, the EMH, and performativity does not make the mapping chain "too reticular" or "too indirect." It makes the mapping chain **systematically biased in a particular direction**. The mapping function, π, itself has had a filter installed: it encodes only some dimensions of T and pushes the rest into the hidden subspace.

This is a particular type of pseudo-small fiber: the readings observed by the auditor, EPS, stock price, credit ratings, are "precise." There is no issue with the precision of the numbers themselves. But the uncertainty structure behind those precise numbers, **which dimensions of T those numbers omit**, remains invisible to the auditor. **Precision conceals systematic omission.**

§6.2 Relation to the Previous Three Chapters

Chapter 19 asks: "Given a price reading, r, can the auditor trace back to the true state, T, that generated the reading?" The answer is "increasingly difficult"—the mapping chain has become reticular, independent components have been diluted, proxy pricing contains no independent information, and scattered readings are hard to integrate.

This chapter asks a more fundamental question: "Even if the auditor locks on, within a given model, a state

compatible with the reading, is that state the **complete** T?" The answer is "no." The mapping function itself encodes only some dimensions of T; what the auditor traces back to is only the projection of T in the visible subspace.

The spatialization diagnosed in Chapter 19 makes $\pi^{-1}(r)$ larger—given a reading r, the set of true states compatible with it becomes larger and larger, so the auditor cannot uniquely determine T. The ideological problem revealed in this chapter makes the candidate state set of the audit procedure dimension-deficient from the outset—the auditor searches within the visible subspace, and dimensions residing in the hidden subspace never enter the candidate set in the first place.

By way of metaphor: Chapter 19 says that "the telescope lacks resolution"; this chapter says that "the telescope can only see a particular band of light." The former is a technical problem, resolution can be improved; the latter is a design problem, the band is chosen by the maker. This distinction is analytically valid, but should not be misread as if "technical problems" and "design problems" can be completely separated. The very tools that improve resolution are themselves designed by specific makers who decide the visible band. It is only that under different analytic conditions one dimension dominates while the other is relatively secondary. At this point, the four-layer diagnostic structure of Chapter 4, §7 (L0-L3) has completed its unfolding in the financial domain—L0, L1, and L3 were founded by Chapter 17 and Chapter 19 at two scales, and the institutional-lock aspect of L2 has been supplied by this chapter; L4 (incentive capture) of the finance section's five-layer progressive structure, together with its micro cases, is taken up by the next chapter (Chapter 17, §0; Chapter 19, §8).

§6.3 The Endogenous Obstacles of Ideological Audit

This leads to a difficulty deeper than the audit difficulties diagnosed in Chapter 19: **the auditor is himself embedded in the same ideological apparatus**.

The auditor in Chapter 19 faces "insufficient capacity"—his tools cannot keep up with the reticularization and dispersion of the mapping chain. But at least he knows that he is facing difficulties—Chapter 19 itself has marked these difficulties out layer by layer: proxy pricing contains no independent information, NBBO integration involves delays, and the global financial cycle makes readings mutually determinative.

The auditor revealed in this chapter faces a "blind zone of vision": he does not know what he cannot see. When the tools used by the auditor, DCF models, PE valuation, risk-adjusted return, are themselves part of the ideological apparatus, the auditor cannot use those tools to detect the blind zones of the tools themselves. This is not "insufficient capacity." It is **"not knowing that one does not know"**—the absence of metacognition.

Within the Fiber Framework, this is the operation of the second derivative of ϕ at the audit level: ideology affects not only the behavior of market participants, the first-order effect, but also the conceptual framework by which auditors understand the market, the second-order effect. When auditors use the EMH as the benchmark for evaluating market efficiency, they have already presupposed that "the market is correct most of the time." This presupposition prevents them from detecting the pricing biases systematically manufactured by the ideological apparatus itself.

Boundary marking: the endogenous obstacles of ideological audit do not mean absolute impossibility. Asness's "Less-Efficient Market Hypothesis" proves that participants within the industry can identify and criticize the ideological apparatus. Asness's doctoral supervisor was Fama; he himself is a core member of the EMH academic lineage; and yet he still advanced an empirical diagnosis contrary to the EMH. This means that "not

knowing that one does not know" is not absolute. It is possible to break through the blind zone of vision. But Asness's diagnosis attributes the difficulty chiefly to the risk profile of the strategy itself (longer and more extreme periods of poor performance, harder perseverance); this chapter's incremental diagnosis concerns how the supply and holding horizon of corrective capital are constrained by the incentive structures at the levels of managers and institutions. Asness can diagnose that "the market has become less efficient," but he cannot by himself alter the incentive conditions that make inefficiency persist. This is precisely the structural power of ideological apparatuses: not to make critique impossible, but to raise the cost of translating critique into institutional change—identifying bias and changing the institutional conditions that produce bias are two different tasks.

Terminology Correspondence Table

The relations between the triple ideological apparatus diagnosed in this chapter and the core concepts of the Fiber Framework and the Macro-Cognitive Accounting Framework (MCAF) are as follows:

Diagnostic object in Chapter 20	Correspondence in the Fiber Framework	Correspondence in the MCAF
Shareholder value maximization (SVO) - institutionally reinforced pricing preference	L2 pseudo-calibration - the filter determines the content of the visible subspace	Institutionalization of ϕ (narrative load) - naturalizing shareholder value maximization as the "only reasonable corporate goal"
Buybacks - arranged around EPS-based assessment, with the improvement in per-share metrics misread as improvement in operating capability	The perceived fiber may be improperly narrowed	Unrecognized financing costs and capability losses may accumulate into cognitive debt (resonating with the shadow-debt mechanism; the gap cannot be read off directly from distribution ratios)
Efficient Market Hypothesis (EMH) - a default coordinate system that needs no argument	Excluding system audit from the default agenda	ϕ'' (second derivative of narrative load) - the meta-rule defining "what counts as a reasonable narrative"
Performativity - the model changes from "description" into "norm"	Established models constrain candidate explanations and alternative metrics	The solidification of ϕ'' at the operational level - the language of the model becomes the only conceptual framework
Superposition of the triple apparatus	Evaluation orientation, threshold for questioning, and operating conventions jointly affect calibration	They constitute institutional conditions for the formation of cognitive debt, from which its magnitude cannot be computed directly

Diagnostic object in Chapter 20	Correspondence in the Fiber Framework	Correspondence in the MCAF
"Breaking the doctrine of financial elitism"	Reverse calibration - institutional dismantling of the fourth link of the "drifting from the real into the virtual" cross-layer transmission chain (ideological masking)	The effect on $\|L\|$ (legitimacy base) awaits examination - political authority compresses the narrative space of ϕ; whether the base is thereby reinforced depends on the persistence and effectiveness of implementation (see the boundary marking in §5.6)

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